

Input Form (IF)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: BLURB | Context: EU/French Pharmaceutical Regulatory NLP — BLURB Assessment

Input Form definition: Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

BLURB is exclusively English-language text [Q1, Q46, Q126], with vocabulary generated from PubMed [Q126] and tokenization optimized for English biomedical terminology [Q29, Q162, Q168]. The deployment operates on French-language regulatory documents with French diacritics and French-specific tokenization needs. No multilingual processing, no French-language variant, and no porting strategy is documented in BLURB. The benchmark’s own argument that vocabulary mismatch causes performance degradation — ‘standard BERT models are forced to divert parametrization capacity and training bandwidth to model biomedical terms using fragmented subwords’ [Q28], with examples like ‘naloxone’ fragmenting into four pieces [Q29] — applies with even greater force across the English↔French boundary. Sequence-length truncations (128/256/512 tokens [Q120, Q125]) were calibrated for PubMed sentence/abstract structure, not for SmPC sections or PIL prose. Web-confirmed French biomedical models (DrBERT, CamemBERT-bio, AliBERT) [WEB-14, WEB-15, WEB-16, WEB-17] exist but are not part of BLURB and have not been fine-tuned on EMA/ANSM regulatory genres. Form mismatch is foundational.

ID	Evidence
Q1	“we compile a comprehensive biomedical NLP benchmark from publicly-available datasets.” (p.1)
Q46	“We focus on PubMed-based biomedical applications, and leave the exploration of the clinical domain, and other high-value verticals to future work.” (p.6)
Q126	“For biomedical domain-specific pretraining, we generate the vocabulary and conduct pretraining using the latest collection of PubMed abstracts: 14 million abstracts, 3.2 billion words, 21 GB.” (p.12)
Q29	“For example, naloxone, a common medical term, is divided into four pieces ([na, ##lo, ##xon, ##e]) by BERT, and acetyltransferase is shattered into seven pieces ([ace, ##ty, ##lt, ##ran, ##sf, ##eras, ##e]) by BERT.” (p.5)
Q162	“A significant advantage in using an in-domain vocabulary is that the input will be shorter in downstream tasks, as shown in Table 8, which makes learning easier.” (p.15)
Q168	“Table 8. Comparison of the average input length in word pieces using general-domain vs in-domain vocabulary.” (p.17)
Q28	“As a result, standard BERT models are forced to divert parametrization capacity and training bandwidth to model biomedical terms using fragmented subwords.” (p.5)
Q120	“For computational efficiency, we use padding or truncation to set the input length to 128 tokens for GAD and 256 tokens for ChemProt and DDI which contain longer input sequences.” (p.12)
Q125	“For computational efficiency, we use padding or truncation to set the input length to 512 tokens.” (p.12)
WEB-14	DrBERT — French biomedical PLM pretrained from scratch (ACL 2023), not part of BLURB (https://arxiv.org/abs/2304.00958)
WEB-15	CamemBERT-bio — French biomedical continual-pretraining model (LREC-COLING 2024) (https://arxiv.org/abs/2306.15550)
WEB-16	AliBERT — French biomedical PLM with Unigram tokenizer (BioNLP 2023) (https://aclanthology.org/2023.bionlp-1.19/)
WEB-17	DrBenchmark — 20-task French biomedical evaluation suite, separate from BLURB and not regulatory-specific (https://aclanthology.org/2024.lrec-main.478/)

Strengths

- BLURB establishes that in-domain vocabulary and tokenization materially affect performance [Q25, Q28, Q160, Q162], providing a transferable methodological lesson the deployment can apply when training a French regulatory tokenizer

- Standard transformer preprocessing scaffolding (special token insertion, padding/truncation, entity dummification) [Q99, Q120, Q183, Q184] is technology-neutral with respect to language and could be reused in a French regulatory NLP pipeline

ID	Evidence
Q25	"A major advantage of domain-specific pretraining from scratch stems from having an in-domain vocabulary." (p.5)
Q28	"As a result, standard BERT models are forced to divert parametrization capacity and training bandwidth to model biomedical terms using fragmented subwords." (p.5)
Q160	"Using the original BERT vocabulary derived from Wikipedia & BookCorpus (by continual pretraining from the original BERT), the results are significantly worse than using an in-domain vocabulary from PubMed." (p.15)
Q162	"A significant advantage in using an in-domain vocabulary is that the input will be shorter in downstream tasks, as shown in Table 8, which makes learning easier." (p.15)
Q99	"An input instance is first processed by a TransformInput module which performs task-specific transformations such as appending special instance marker (e.g., [CLS]) or dummifying entity mentions for relation extraction." (p.10)
Q120	"For computational efficiency, we use padding or truncation to set the input length to 128 tokens for GAD and 256 tokens for ChemProt and DDI which contain longer input sequences." (p.12)
Q183	"With entity dummification, the entity mentions in question are anonymized using entity type tags such as \$DRUG or \$GENE." (p.20)
Q184	"With entity marker, special tags marking the start and end of an entity are appended to the entity mentions in question." (p.20)

Checklist

Checklist item definitions:

ID	Assessment Question
IF-1	Compare signal distributions between source and target
IF-2	Check if regional infrastructure supports data capture specs
IF-3	Identify domain-specific form differences
IF-4	Document form mismatches harming external validity

Checklist responses:

IF-1: Signal distributions are mismatched: BLURB inputs are English PubMed text tokenized with a PubMed-derived vocabulary [Q126], whereas the deployment requires French regulatory text with French diacritics (é, è, ê, ç, œ) and regulatory-specific tokens (INNs, ATC codes, posology phrases). The benchmark's own demonstration that vocabulary fragmentation harms performance [Q28, Q29] applies to this mismatch.

IF-2: Infrastructure is not the constraint — the deployment population has full enterprise infrastructure. The form mismatch is at the linguistic/encoding level, not the capture level.

IF-3: Yes: regulatory documents have template-driven structure (numbered SmPC sections, PIL headings) that BLURB's flat-abstract preprocessing [Q84] does not model. Sequence-length caps of 128/256/512 tokens [Q120, Q125] may also truncate long SmPC sections or cross-section equivalence pairs.

IF-4: External validity is severely harmed: the benchmark cannot evaluate model behavior on French text, French regulatory tokenization, or template-structured documents — precisely the input form the deployment must handle.

ID	Evidence
Q126	"For biomedical domain-specific pretraining, we generate the vocabulary and conduct pretraining using the latest collection of PubMed abstracts: 14 million abstracts, 3.2 billion words, 21 GB." (p.12)
Q28	"As a result, standard BERT models are forced to divert parametrization capacity and training bandwidth to model biomedical terms using fragmented subwords." (p.5)
Q29	"For example, naloxone, a common medical term, is divided into four pieces ([na, ##lo, ##xon, ##e]) by BERT, and acetyltransferase is shattered into seven pieces ([ace, ##ty, ##lt, ##ran, ##sf, ##eras, ##e]) by BERT." (p.5)
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Q84	"Though the original dataset provided sentence level annotation, we follow the common practice and evaluate on the abstract level [19, 60]." (p.9)
Q120	"For computational efficiency, we use padding or truncation to set the input length to 128 tokens for GAD and 256 tokens for ChemProt and DDI which contain longer input sequences." (p.12)
Q125	"For computational efficiency, we use padding or truncation to set the input length to 512 tokens." (p.12)
Q1	"we compile a comprehensive biomedical NLP benchmark from publicly-available datasets." (p.1)
WEB-17	DrBenchmark — 20-task French biomedical evaluation suite, separate from BLURB and not regulatory-specific (https://aclanthology.org/2024.lrec-main.478/)

Information Gaps

- Whether SmPC/PIL section length distributions exceed BLURB's 512-token cap in practice

Requires Expert Verification

- Optimal French regulatory tokenizer construction strategy (vocabulary scope, treatment of INNs/ATC codes as multi-word units)

Remediation

Gap 1: BLURB is exclusively English with PubMed-derived tokenization [Q126]; the deployment requires French tokenization with regulatory vocabulary support.

Recommendation: Adopt a French biomedical PLM (DrBERT, CamemBERT-bio, or AliBERT [WEB-14, WEB-15, WEB-16]) as the encoder and extend its tokenizer with regulatory-specific tokens (INNs, ATC codes, French posology multi-word units), applying BLURB's own in-domain-vocabulary lessons [Q25, Q28].

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